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Assignment Logic Design

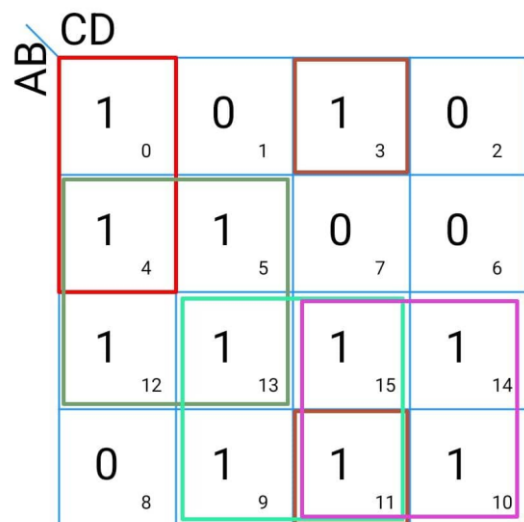
B2IT

Design a logic circuit and function for the following:

Truth table

A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Using a Karnaugh map:



The Function is: $A'C'D + B'CD + BC' + AD + AC$

using sum of products SOP, we conclude the following:

$$\begin{aligned} &A'.B'.C'.D' + A'.B'.C.D + A'.B.C'.D' + A'.B.C'.D \\ &+ A.B'.C'.D + A.B'.C.D' + A.B'.C.D + A.B.C'.D' \\ &+ A.B.C'.D + A.B.C.D' + A.B.C.D \end{aligned}$$

Then we simplify it to the following:

$$A'C'D + B'CD + BC' + AD + AC$$

